

ComplyPilot

An agent that takes an SME's messy paper invoices, converts them into the new legally-mandated tax invoice format, and then *files them itself* by driving the government portal on screen.

One-line pitch: Sri Lanka made a new invoice format legally binding on 1 July 2026 and is switching tax filing to a Web API. Hundreds of thousands of SMEs run on notebooks and Excel and cannot comply. ComplyPilot is the bridge — it reads what they already have, fixes what's non-compliant, and submits it, with a human approving only the ambiguous cases.

1. The problem (and why it's urgent right now)

- A revised, **legally binding tax invoice format** applies to all VAT-registered persons from **1 July 2026** (Gazette Extraordinary No. 2481/22, Circular SEC/2026/E/03). The old format is withdrawn.
- It includes structural requirements SMEs cannot produce by hand — e.g. a unique serial in the format `YYMM_QQQQ_XXXX`, exact TIN/name/address matching the VAT certificate, and three LKR figures to two decimals.
- In parallel, the IRD is rolling out a **National e-Invoicing System** that transmits invoice data to RAMIS over a secure Web API — phased from export-oriented enterprises to all VAT-registered persons.
- Published analysis of SME readiness is blunt: a significant proportion of Sri Lankan SMEs operate on basic accounting tools or manual invoicing, keep no structured data fields or classification codes, and have no dedicated finance or IT staff.

The gap: ERP vendors serve the top of the market. Everyone else is being asked to become structured-data compliant with no tooling, no budget, and no API access to their own filings.

2. What we build

Layer 1 — Ingest. Phone photos of receipt books, scanned PDFs, WhatsApp images, Excel sheets, handwritten ledgers. A vision-language agent extracts line items, parties, amounts and dates.

Layer 2 — Compliance engine. Extracted data is mapped against a machine-readable rules pack for the gazetted format. Every field is scored: *compliant* / *auto-fixable* / *needs human*. The agent generates valid serials, normalises units and currency, and drops anything out-of-scope that shouldn't appear on a tax invoice. Only genuine ambiguities go to a human review queue.

Layer 3 — Filing (the hard part). Government portals have no public API for small filers. So the agent doesn't integrate — it **operates the portal**. A GUI agent reads the screen, fills the forms, hands off to the human for OTP/CAPTCHA, recovers from session timeouts, and records a video + structured audit trail of every action it took.

Why layer 3 is the differentiator

Every team at this hackathon will build something that extracts data. Almost none will close the loop to actual submission, because on most stacks a GUI agent is prohibitively expensive. Qwen3.7-Plus blends GUI and CLI operation inside a single agent loop, which makes this economically viable — and it is the single most memorable thing a judge will watch that day.

3. Architecture & Alibaba Cloud stack

“Use of Alibaba Cloud products” is a separately scored judging category. Every component below is deliberately native.

COMPONENT	SERVICE	ROLE
Document understanding	Qwen3.7-Plus (VL) via Model Studio	OCR, layout parsing, handwriting, line-item extraction
GUI / portal agent	Qwen3.7-Plus — unified GUI + CLI loop	Drives the filing portal, verifies each step, recovers from failures
Orchestration	Qwen3.7-Max	Long-horizon planning, tool calls, human-in-loop checkpoints
Rules pack + retrieval	AnalyticDB (vector + relational)	Gazette clauses, field specs, per-sector guidance; citation source for every decision
Batch pipeline	Function Compute	Fan-out extraction across a document dump
Hosting	ECS + OSS	Backend, review UI, document + audit-trail storage

Submission requirement, do not skip: a short screen recording proving the backend actually runs on Alibaba Cloud, plus an architecture diagram, plus a ~3 minute demo video. These are graded artefacts, not paperwork.

4. Why this wins against the field

JUDGING CRITERION	HOW WE HIT IT
Technical depth & engineering	Two hard problems, not one: multimodal extraction <i>and</i> stateful GUI automation with recovery and audit trails
Use of Alibaba Cloud products	Six native services; the core capability (cheap GUI agent) is only viable on this stack
Real-world value & impact	A live 2026 legal mandate with a documented SME readiness gap — the problem needs zero explaining
Innovation	Nobody automates the <i>submission</i> step. Everyone automates extraction.
Presentation	Shoebox of receipts in, filed returns out. One visual, 90 seconds.

Precedent: Alibaba Cloud’s Malaysia 2025 winners were InsureScan (1st, document + damage analysis for insurance claims), a freelancer tax-filing assistant using OCR and Qwen (2nd), and FundSight AI for SME grant matching (3rd). The winning pattern across every regional event is the same: **a boring, painful, regulated vertical workflow — not a general chatbot**. This is that pattern, one level harder.

5. The 3-minute demo script

1. **0:00-0:20** — Hold up a real receipt book. State the date the law changed and what happens to a non-compliant invoice.
2. **0:20-1:10** — Drop ~40 phone photos in. Watch them become structured, compliant invoices. Three land in the review queue; a human resolves one on screen.
3. **1:10-2:10** — The agent opens the portal and files. Show the audit-trail replay. (Pre-record this segment. Do not run it live.)
4. **2:10-2:40** — The numbers slide: manual minutes per invoice vs. ours, and error rate on a held-out set.
5. **2:40-3:00** — Architecture diagram + the rules pack is swappable, so this ports to Pakistan, UAE, Indonesia by replacing one file.

6. Scope & split of work

Must ship (MVP)

- Extraction on one document type (photographed receipt book)
- Rules pack for the mandatory field list
- Serial generation + validation
- Review queue UI with approve/edit
- GUI agent filing against a mock portal
- Deployed on ECS, recording captured

Only if time remains

- Excel / CSV ingest path
- Multi-language UI (Sinhala / Tamil)
- Credit & debit note handling
- Bulk reconciliation against bank records
- Second-country rules pack as proof of portability

Scoping discipline: the most common way teams lose is over-scoping. One clean end-to-end path beats five half-working features. If the GUI agent is fighting us at the 60% mark, ship extraction + compliance + a mock-portal filing and spend the remaining time on the demo video.

7. Risks

RISK	MITIGATION
GUI agent is brittle live	Pre-record that segment; build against a mock portal we control, not the live government one

Handwriting extraction accuracy	Confidence thresholds route low-confidence fields to the human queue — and that queue is a feature, not a failure
Judges see it as “just OCR”	Lead the pitch with the filing loop and the audit trail, not the extraction
Legal/ethical concern about automating a government portal	Human approval before every submission; full immutable audit trail; frame as assisted filing, never silent filing

Sources. Gazette Extraordinary No. 2481/22 (27 Mar 2026) and IRD Circular SEC/2026/E/03 (20 May 2026), via LookupTax Sri Lanka VAT guidelines · “VAT Invoice Reform 2026: A necessary step forward but are SMEs ready?”, Daily FT · “Sri Lanka Launches National E-Invoicing System for VAT Under 2026 Budget”, VATupdate · Alibaba Cloud Model Studio and Qwen3.7-Plus / Qwen3.7-Max release notes · “Inaugural Alibaba Cloud AI Hackathon Malaysia”, Alibaba Cloud Community · Global AI Hackathon Series with Qwen Cloud submission requirements, Devpost.

Working name only. Verify the exact rules-pack details against the current IRD circular before the pitch — regulations have already shifted dates once.